



Volumetric Error Compensation in CNC Machine Tools and CMS

Hocken R.J., P.H. Pereira (2012)
Coordinate Measuring Machines and Systems
Chapter 12



The student knows:

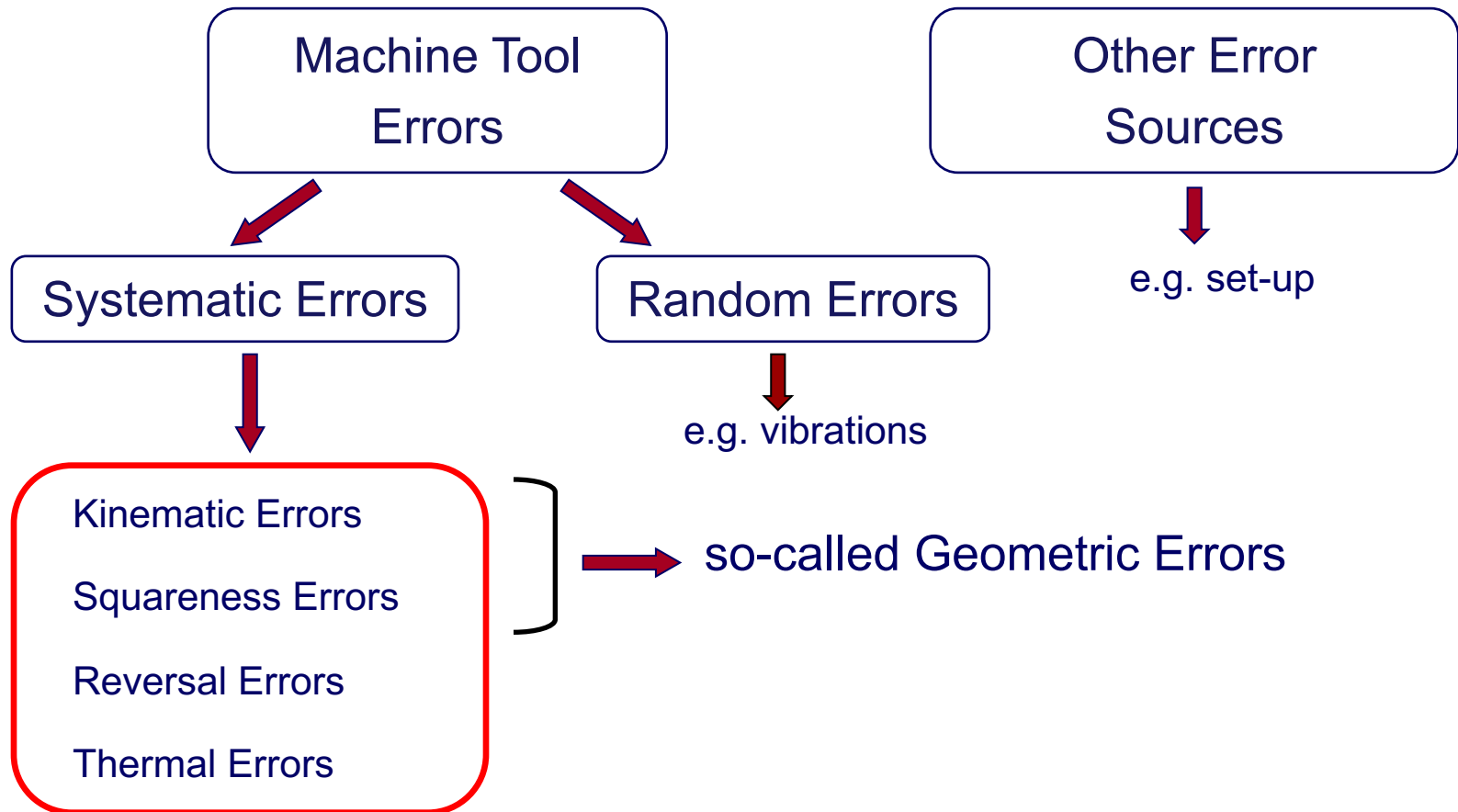
- What the volumetric error of a CMM/CMS/MT is
- How to model the volumetric error
- How to compensate the volumetric error





MACHINE TOOL ERROR SOURCES

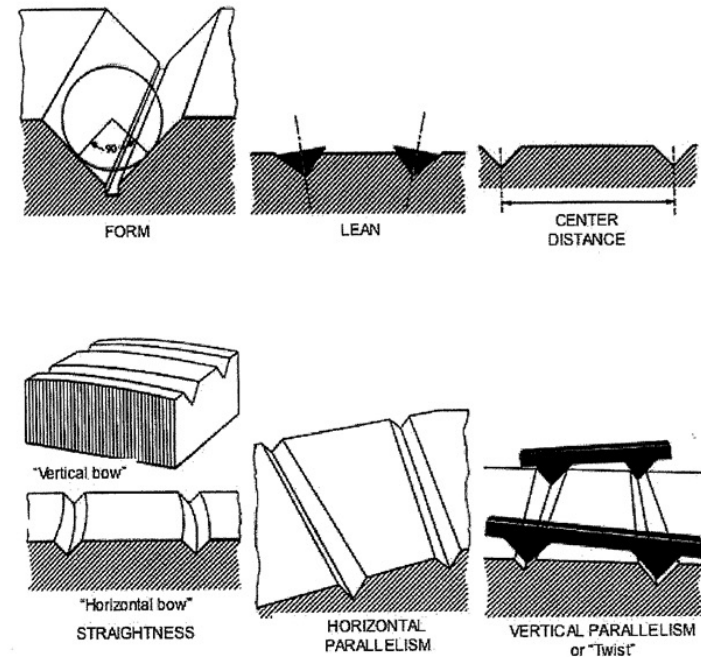
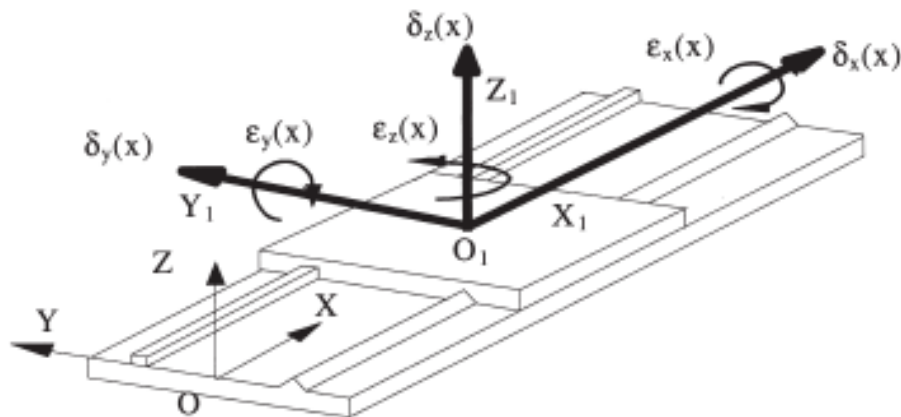
Machine tool performance and accuracy is affected by various error sources that influence the accuracy of the machined parts. They are mainly:





MACHINE TOOL ERROR SOURCES

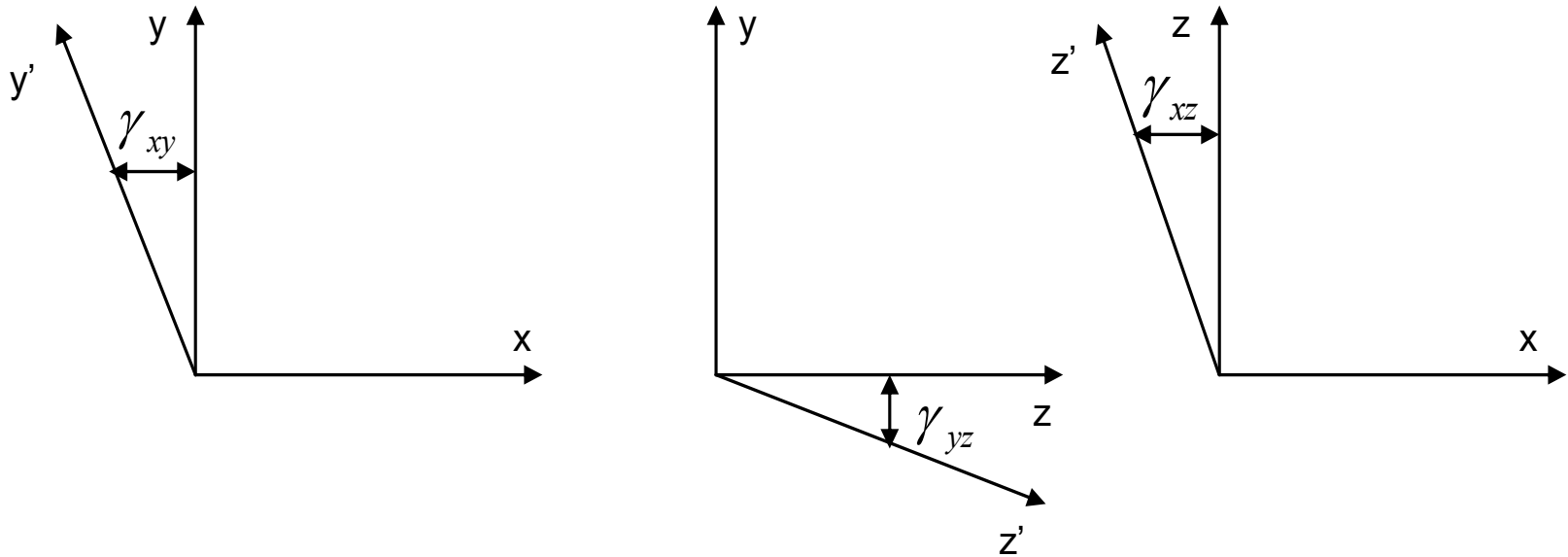
Kinematic Errors: During the positioning of each axis it is possible to have up to six errors, three linear errors (a linear positioning error and two straightness errors) and three rotation errors (roll, pitch and yaw).





MACHINE TOOL ERROR SOURCES

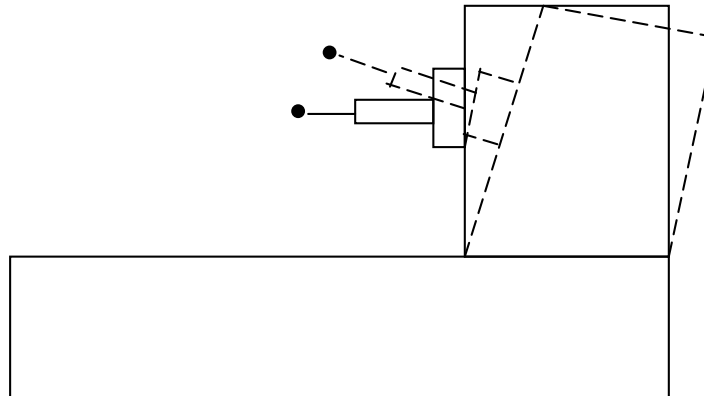
Squareness Errors: Perpendicularity errors between each couple of axes, mainly due to assembly.





MACHINE TOOL ERROR SOURCES

Thermal Errors: Structural deformation due to thermal effects.

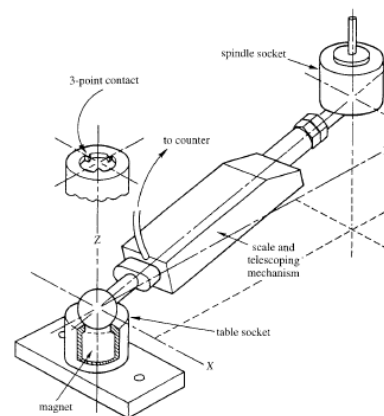
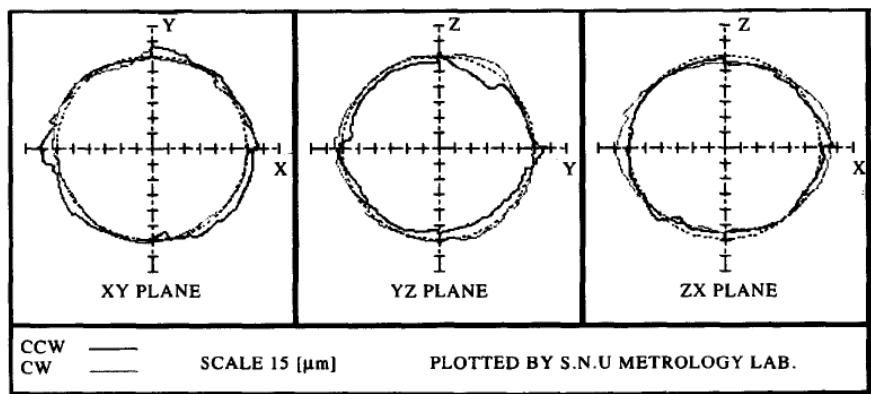




MACHINE TOOL ERROR SOURCES

7

Reversal Errors: Errors due to many reasons during the change in the direction of the motion of a linear axis.

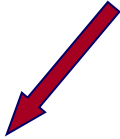




Kinematic, squareness, and thermal errors could be modeled, therefore they could be predicted, measured and compensated.

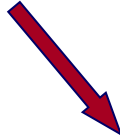
These models define the error of the positioning of the tool with respect to the workpiece inside the machine tool working envelope, the so-called volumetric error of the machine tool.

It is possible to consider different situations (hypotheses):



Machine tool as an
assembly of rigid bodies

- 
- Geometric Errors



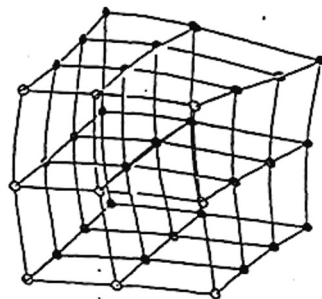
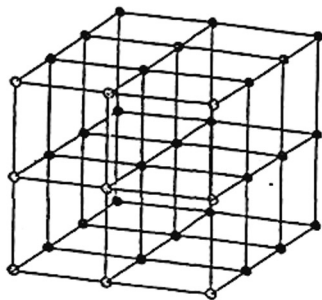
Machine tool as an assembly
of quasi-rigid bodies

- 
- Geometric Errors
 - Thermal Errors



VOLUMETRIC ERROR

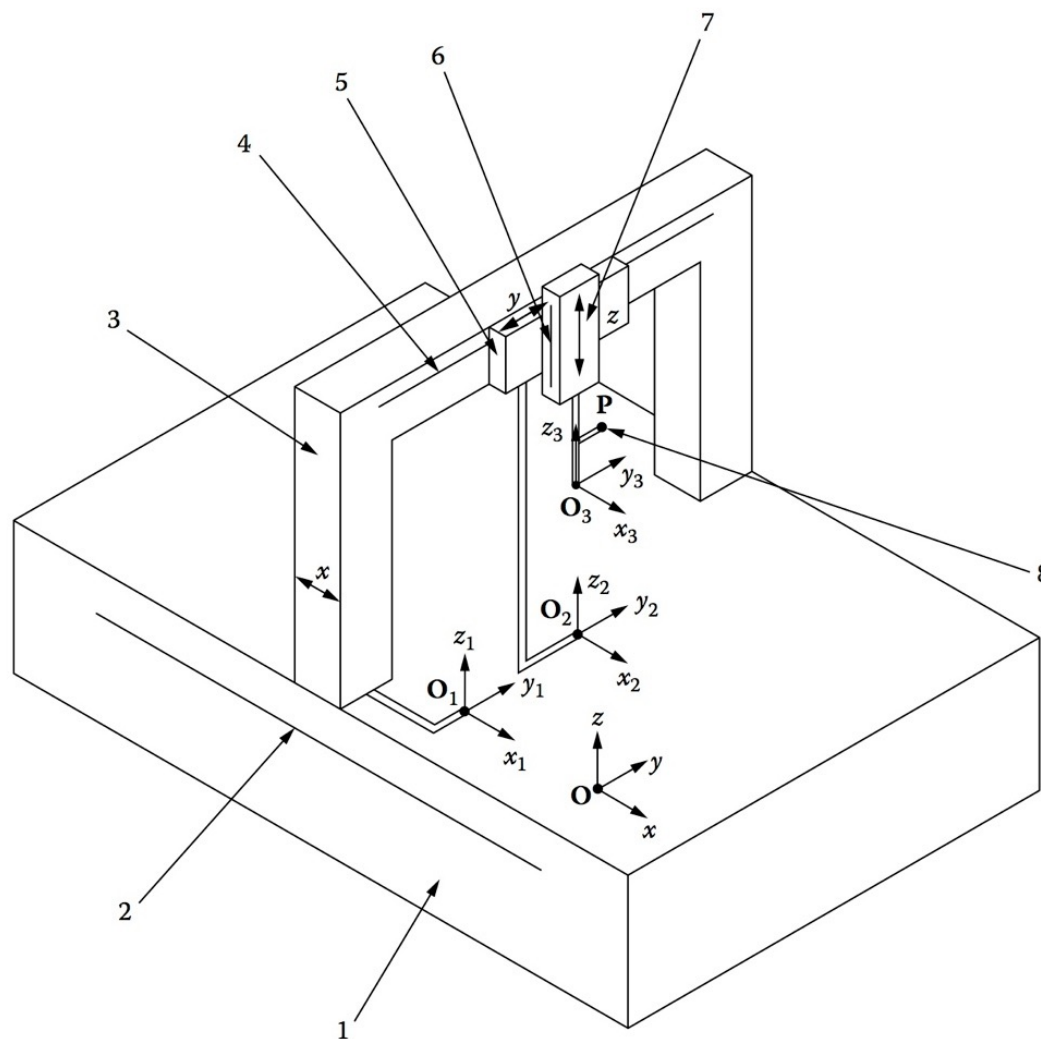
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	3-axis Machine Tool	5-axis Machine Tool
Geometric Errors	21-error model	configuration dependent (>40 errors)
Geometric and Thermal Errors	configuration dependent (>40 errors)	very complex



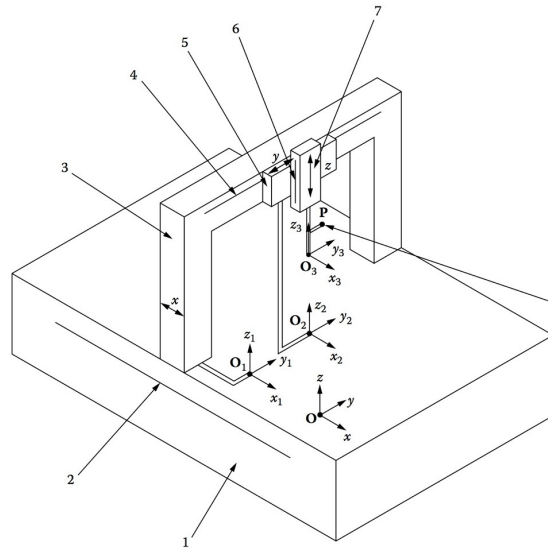
Cartesian MT schema: FXYZ





VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{OO}_1 = \mathbf{X} = \begin{pmatrix} X + \delta_x(X) \\ \delta_y(X) \\ \delta_z(X) \end{pmatrix}$$



$$\mathbf{R}(x) = \begin{pmatrix} \cos(\varepsilon_z(X)) & \sin(\varepsilon_z(X)) & 0 \\ -\sin(\varepsilon_z(X)) & \cos(\varepsilon_z(X)) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\varepsilon_y(X)) & 0 & -\sin(\varepsilon_y(X)) \\ 0 & 1 & 0 \\ \sin(\varepsilon_y(X)) & 0 & \cos(\varepsilon_y(X)) \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\varepsilon_x(X)) & \sin(\varepsilon_x(X)) \\ 0 & -\sin(\varepsilon_x(X)) & \cos(\varepsilon_x(X)) \end{pmatrix}$$

$$\mathbf{R}(x) = \begin{pmatrix} 1 & \varepsilon_z(X) & 0 \\ -\varepsilon_z(X) & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -\varepsilon_y(X) \\ 0 & 1 & 0 \\ \varepsilon_y(X) & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \varepsilon_x(X) \\ 0 & -\varepsilon_x(X) & 1 \end{pmatrix} = \begin{pmatrix} 1 & \varepsilon_z(X) & -\varepsilon_y(X) \\ -\varepsilon_z(X) & 1 & \varepsilon_x(X) \\ \varepsilon_y(X) & -\varepsilon_x(X) & 1 \end{pmatrix}$$

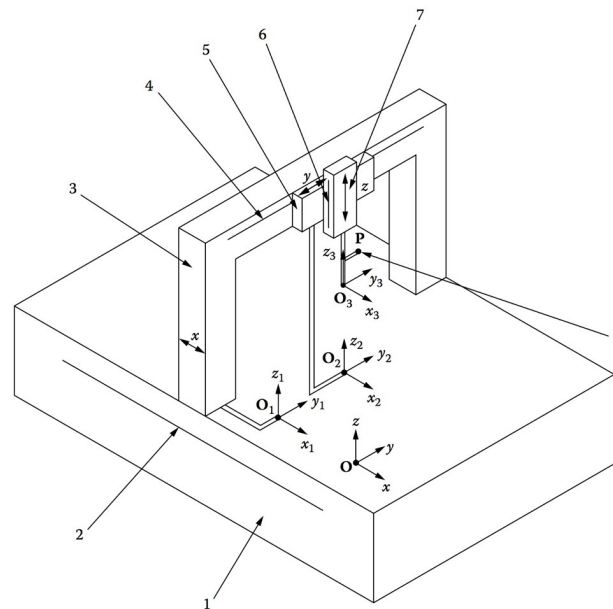


VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{OP} = \mathbf{P} = \begin{pmatrix} X_P & Y_P & Z_P \end{pmatrix}^T$$

$$\mathbf{O}_1\mathbf{P} = \mathbf{P}_1 = \begin{pmatrix} X_{P1} & Y_{P1} & Z_{P1} \end{pmatrix}^T$$

$$\mathbf{P} = \mathbf{X} + \mathbf{R}^{-1}(x)\mathbf{P}_1$$



$$\begin{pmatrix} X_P \\ Y_P \\ Z_P \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & \varepsilon_z(X) & -\varepsilon_y(X) & X + \delta_x(X) \\ -\varepsilon_z(X) & 1 & \varepsilon_x(X) & \delta_y(X) \\ \varepsilon_y(X) & -\varepsilon_x(X) & 1 & \delta_z(X) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_{P1} \\ Y_{P1} \\ Z_{P1} \\ 1 \end{pmatrix}$$

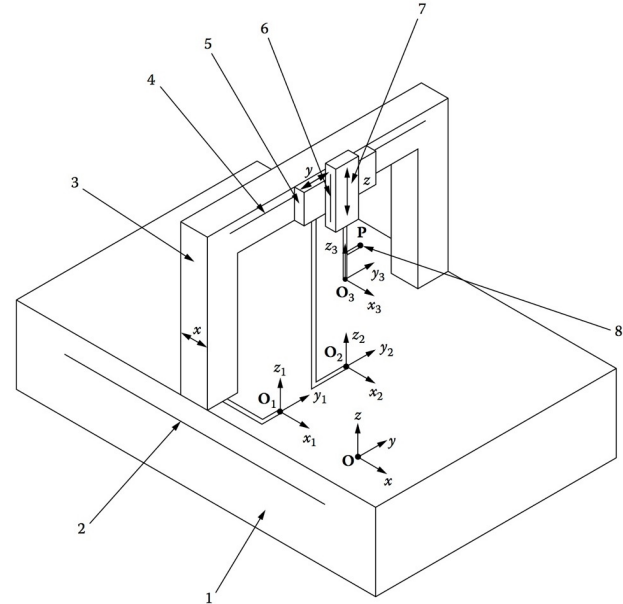


VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{O}_1 \mathbf{P} = \mathbf{P} = \begin{pmatrix} X_{P1} & Y_{P1} & Z_{P1} \end{pmatrix}^T$$

$$\mathbf{O}_2 \mathbf{P} = \mathbf{P}_2 = \begin{pmatrix} X_{P2} & Y_{P2} & Z_{P2} \end{pmatrix}^T$$

$$\mathbf{P}_1 = \mathbf{Y} + \mathbf{R}^{-1}(y) \mathbf{P}_2$$



$$\begin{pmatrix} X_{P1} \\ Y_{P1} \\ Z_{P1} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\varepsilon_z(Y) & \varepsilon_y(Y) & \delta_x(Y) - Y\alpha_{xy} \\ \varepsilon_z(Y) & 1 & -\varepsilon_x(Y) & Y + \delta_y(Y) \\ -\varepsilon_y(Y) & \varepsilon_x(Y) & 1 & \delta_z(Y) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_{P2} \\ Y_{P2} \\ Z_{P2} \\ 1 \end{pmatrix}$$

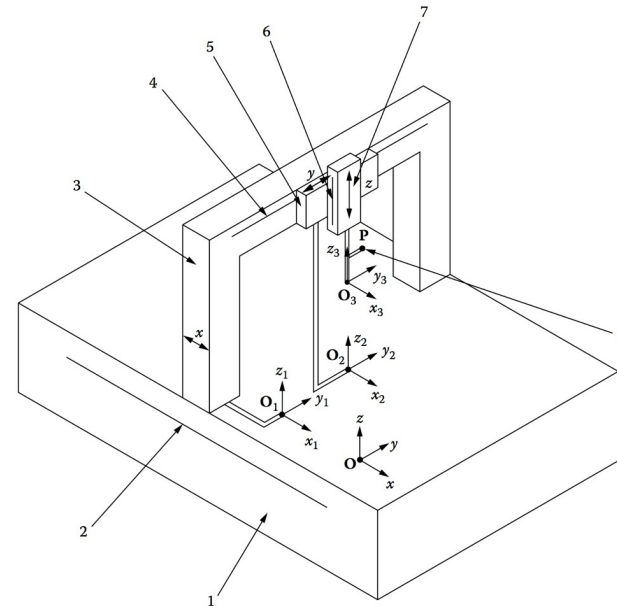


VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{O}_2 \mathbf{P} = \mathbf{P} = \begin{pmatrix} X_{P2} & Y_{P2} & Z_{P2} \end{pmatrix}^T$$

$$\mathbf{O}_3 \mathbf{P} = \mathbf{P}_3 = \begin{pmatrix} X_{P3} & Y_{P3} & Z_{P3} \end{pmatrix}^T$$

$$\mathbf{P}_2 = \mathbf{Z} + \mathbf{R}^{-1}(z) \mathbf{P}_3$$



$$\begin{pmatrix} X_{P2} \\ Y_{P2} \\ Z_{P2} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\varepsilon_z(Z) & \varepsilon_y(Z) & \delta_x(Z) - Z\alpha_{zx} \\ \varepsilon_z(Z) & 1 & -\varepsilon_x(Z) & \delta_y(Z) - Z\alpha_{yz} \\ -\varepsilon_y(Z) & \varepsilon_x(Z) & 1 & Z + \delta_z(Z) \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_{P3} \\ Y_{P3} \\ Z_{P3} \\ 1 \end{pmatrix}$$

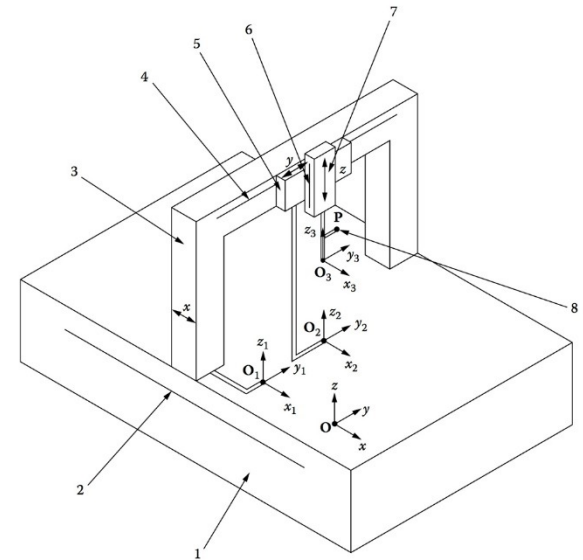


VOLUMETRIC ERROR: 3-AXIS CNC MACHINE TOOL

$$\mathbf{OP} = \mathbf{P} = \begin{pmatrix} X_P & Y_P & Z_P \end{pmatrix}^T$$

$$\mathbf{O}_3\mathbf{P} = \mathbf{P}_3 = \begin{pmatrix} X_{P3} & Y_{P3} & Z_{P3} \end{pmatrix}^T$$

$$\mathbf{P} = \mathbf{X} + \mathbf{R}^{-1}(x) \left\{ \mathbf{Y} + \mathbf{R}^{-1}(y) \left[\mathbf{Z} + \mathbf{R}^{-1}(z) \mathbf{P}_3 \right] \right\}$$



Ideal Machine Tool:

$$\begin{cases} X_P = X + X_{P3} \\ Y_P = Y + Y_{P3} \\ Z_P = Z + Z_{P3} \end{cases}$$

Real Machine Tool:

$$\begin{cases} \Delta X = X_P - (X + X_{P3}) \\ \Delta Y = Y_P - (Y + Y_{P3}) \\ \Delta Z = Z_P - (Z + Z_{P3}) \end{cases}$$



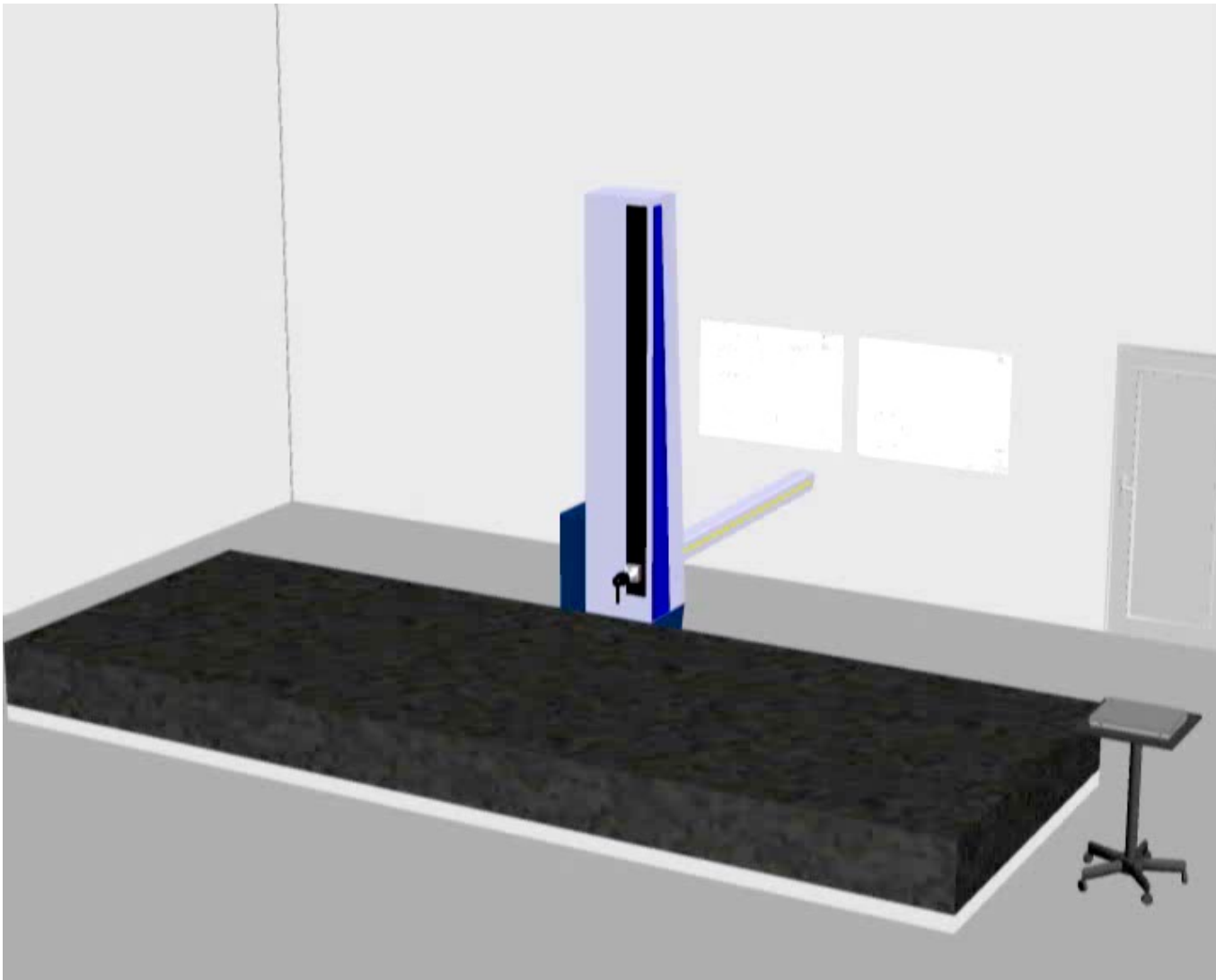
The final result is:

$$\begin{aligned}\Delta X &= \delta_x(X) + \delta_x(Y) + \delta_x(Z) - Y\alpha_{xy} - Z\alpha_{xz} - Y\varepsilon_z(X) + Z[\varepsilon_y(X) + \varepsilon_y(Y)] + \\ &\quad - Y_{P3}[\varepsilon_z(X) + \varepsilon_z(Y) + \varepsilon_z(Z)] + Z_{P3}[\varepsilon_y(X) + \varepsilon_y(Y) + \varepsilon_y(Z)] \\ \Delta Y &= \delta_y(X) + \delta_y(Y) + \delta_y(Z) - Z\alpha_{yz} - Z[\varepsilon_x(X) + \varepsilon_x(Y)] + \\ &\quad + X_{P3}[\varepsilon_z(X) + \varepsilon_z(Y) + \varepsilon_z(Z)] - Z_{P3}[\varepsilon_x(X) + \varepsilon_x(Y) + \varepsilon_x(Z)] \\ \Delta Z &= \delta_z(X) + \delta_z(Y) + \delta_z(Z) + Y\varepsilon_x(X) - X_{P3}[\varepsilon_y(X) + \varepsilon_y(Y) + \varepsilon_y(Z)] \\ &\quad + Y_{P3}[\varepsilon_x(X) + \varepsilon_x(Y) + \varepsilon_x(Z)]\end{aligned}$$



VOLUMETRIC ERRORS MEASUREMENT

17



<https://youtu.be/v4eb0hotUiw>



It adopts a laser interferometer to measure a distance, and simultaneously measures two angles, from which the position in space of the target is derived.





iGPS (INDOOR GLOBAL POSITIONING SYSTEM)

A constellation of emitters (passive) sends a signal and a sensor (active), by triangulation of these emitters, identifies its position.





SOFTWARE VOLUMETRIC ERROR COMPENSATION

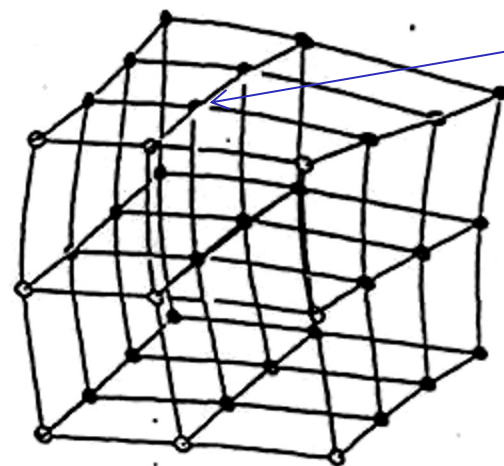
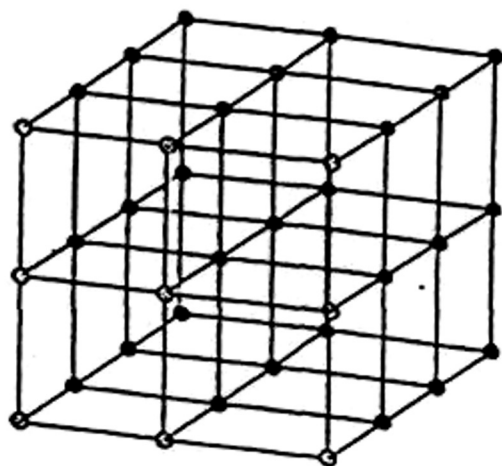
At present CNCs allows three possible representations of the model of the volumetric error :

- error grid;
- error table;
- coefficient table.

SOFTWARE VOLUMETRIC ERROR COMPENSATION

Error Grid: the working envelope of the machine tool is discretized by a regular grid, and for each node of the grid the three components of the error are memorized.

If we consider n point for each axis, we'll need $3n^3$ error components. Compensation in an intermediate point will be done by linear interpolation.



$$\begin{cases} \Delta X_i \\ \Delta Y_i \\ \Delta Z_i \end{cases}$$



SOFTWARE VOLUMETRIC ERROR COMPENSATION

Error Table: For each axis, discretised in n points, we keep memory of the corresponding value of each kinematic error function. The value of a kinematic error function in an intermediate point of the axis will be computed by linear interpolation.

Starting from these data, the volumetric error is computed from its model.

X Axis	Kinematic errors as function of axis position			
Position	F1	F2	...	F18
1	...			
2				
3				
...				
n-1				
n				...



SOFTWARE VOLUMETRIC ERROR COMPENSATION

Coefficient Table: If we know the model of each kinematic error function, it is possible to memorize only the coefficients of the model. Usually the model could be an interpolating cube polynomial function.

$$f1 : a_1 + b_1x + c_1x^2 + d_1x^3$$

$$f2 : a_2 + b_2x + c_2x^2 + d_2x^3 \longrightarrow [18 \cdot (o + 1)]$$

.....

Memorized data

$$f18 : a_{18} + b_{18}x + c_{18}x^2 + d_{18}x^3$$



SOFTWARE VOLUMETRIC ERROR COMPENSATION

Error Grid

- Allows a complete representation of the volumetric error in the whole measuring volume of the machine.
- Only few CNCs allow it.

Error Table

- Very diffused, as it is coherent with the commonly adopted methods for the measurements of the volumetric error.
- Difficulties in the identification of the interactions between the various axes.

Coefficient Table

- Potentially effective.
- Very complicated (hard to manage with current CNCs).



VOLUMETRIC ERROR COMPENSATION

Once the volumetric error is known, there are three ways to consider it to improve the precision of the machine tool:

1. Hardware correction (components reworking);
2. Software compensation of measured position (open loop compensation, it applies only to CMS):

$$\left(X + \Delta X; \quad Y + \Delta Y; \quad Z + \Delta Z \right)$$

3. Software volumetric error compensation according to the following schema (closed loop compensation):

